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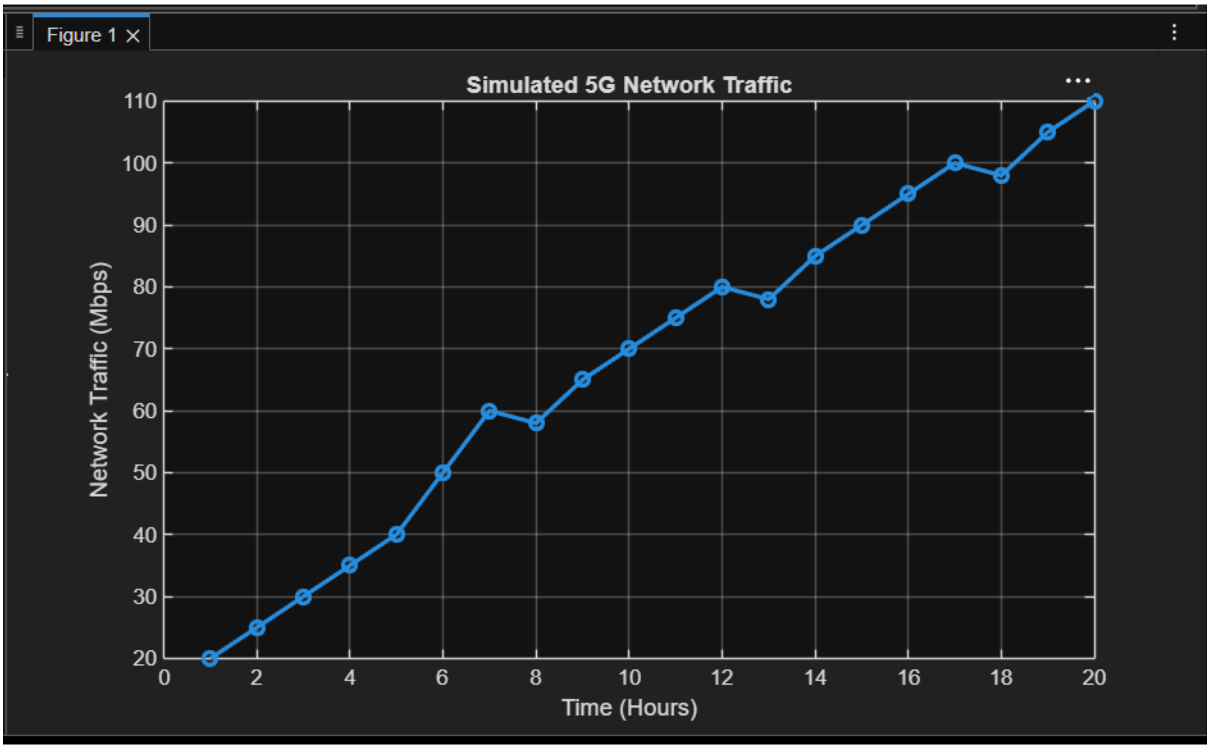
15. Experiment: AI-based Traffic

Prediction for 5G Networks Objective1:

Code:

```
clc;

clear;

close

all;

% Simulated 5G Network Traffic (Mbps)
time = 1:20;
traffic = [20 25 30 35 40 50 60 58 65 70 ...
 75 80 78 85 90 95 100 98
                    105 110];

figure;
      plot(time, traffic, '-o',
             'LineWidth', 2);
grid on;

xlabel('Time (Hours)');

ylabel('Network Traffic

(Mbps)'); title('Simulated 5G

Network Traffic');
```

Output:



Objective 2:

Code:

```
clc;

clear;

close

all;

% Simulated

Traffic Data

time = 1:20;

traffic = [20 25 30 35 40 50 60 58 65 70 ...

    75 80 78 85 90 95 100 98 105 110];
% Linear Regression

Model model =

polyfit(time, traffic,

1);

% Predicted Traffic
```

```
predicted = polyval(model, time);
% Plot Actual and Predicted Traffi
```

figure;

plot(time, traffic, 'bo-',

'LineWidth', 2); hold on;

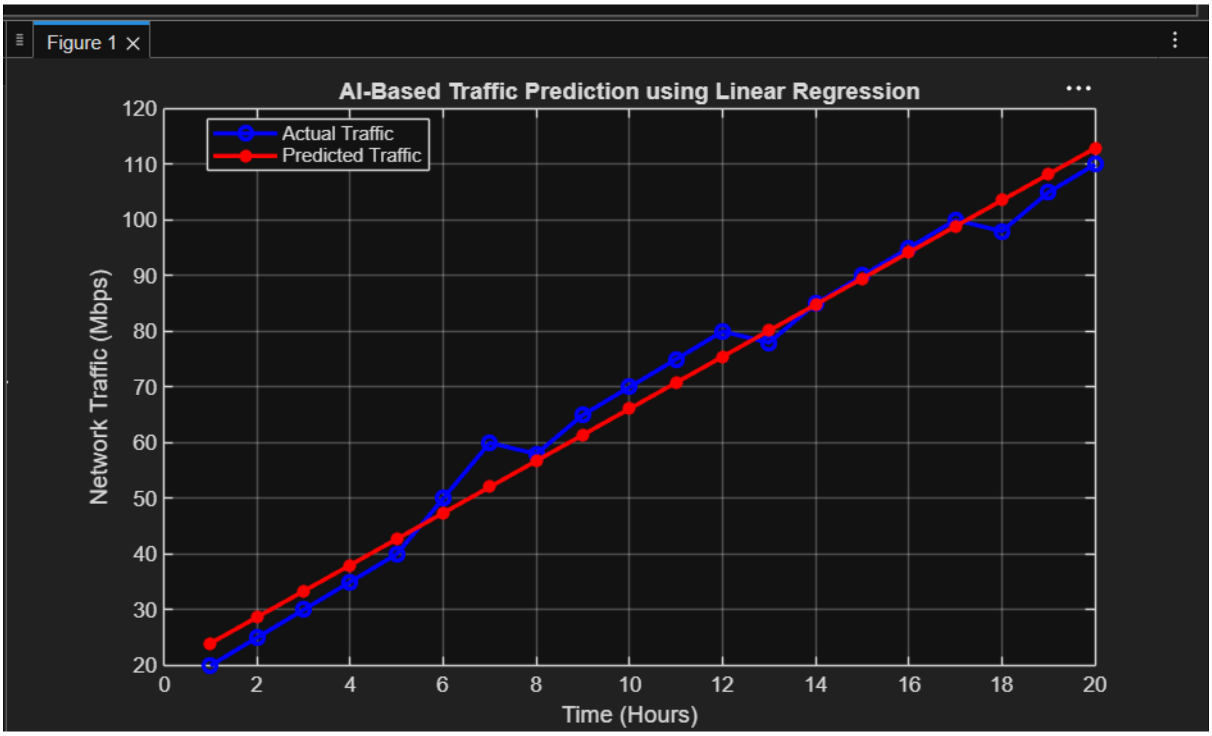
plot(time, predicted, 'r*-', 'LineWidth', 2);
grid on;
xlabel('Time (Hours)');

ylabel('Network Traffic (Mbps)');

title('AI-Based Traffic Prediction using Linear Regression');

legend('Actual Traffic', 'Predicted Traffic',

'Location', 'best'); Output:



Objecti

ve 3:

Code:

```
clc;

clear;

close

all;

% Simulated

Traffic Data

time = 1:20;

traffic = [20 25 30 35 40 50 60 58 65 70 ...
      75 80 78 85 90 95 100 98 105 110];
% Linear Regression

Model model =

polyfit(time, traffic,

1);

% Predicted Traffic

predicted = polyval(model, time);
% Calculate

Prediction Error error

= traffic - predicted;

% Calculate RMSE

RMSE = sqrt(mean(error.^2));
disp(['Root Mean Square Error (RMSE) = ', num2str(RMSE)]);
% Plot

Results

figure;

% Actual vs Predicted

Traffic subplot(2,1,1);

plot(time, traffic, 'bo-',
```

```
'LineWidth', 2); hold on;
```

```
plot(time, predicted, 'r*-',
'LineWidth', 2); grid on;
xlabel('Time
(Hours)');
ylabel('Traffic
(Mbps)');
title('Actual vs Predicted Traffic');
legend('Actual', 'Predicted', 'Location', 'best');
% Prediction
Error
subplot(2,1,2)
; bar(time,
error); grid
on;
xlabel('Time (Hours)');
ylabel('Prediction Error
(Mbps)'); title('Prediction
Error');
% Display Results
disp('Time  Actual
Predicted');
disp([time' traffic'
predicted']); output:
```

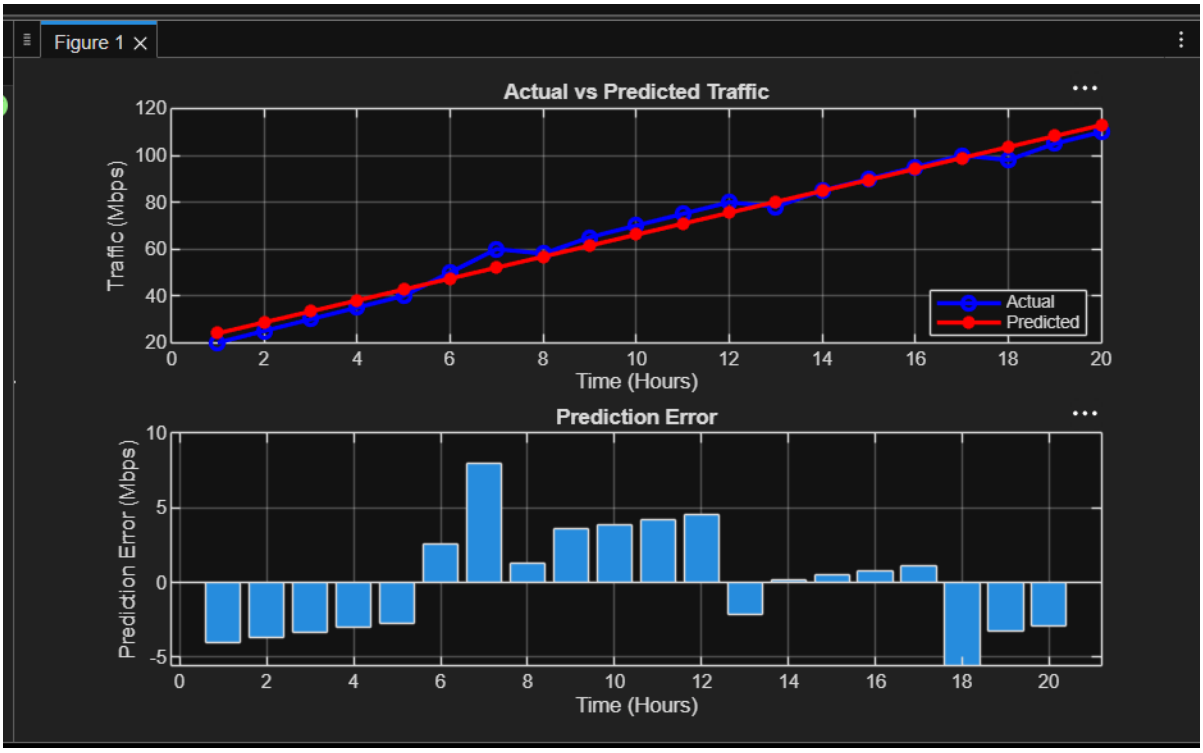


Figure 1
Actual vs Predicted Traffic
Traffic (Mbps)
Time (Hours)
Actual
Predicted
Prediction Error
Prediction Error (Mbps)
Time (Hours)